

A3 Robustness Checks

A3.1 Alternative Fixed Effects

Table A7: Effect of LLM exposure on firm-level job outcomes, time fixed effects

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
LLM exposure x Post	-0.00002 (0.001)	-0.004** (0.001)	0.002 (0.002)	-0.002*** (0.0002)	117.995*** (45.546)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.982	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, using year-by-month fixed effects instead of industry-by-period fixed effects. All other definitions are as in Table 3.

Table A8: Effect of LLM exposure on firm-level job listings, time fixed effects

	Job postings
LLM exposure x Post	-0.018*** (0.002)
Firm fixed-effects	Yes
Time fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.519

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity, using year-by-month fixed effects instead of industry-by-period fixed effects. All other definitions are as in Table A1.

Table A9: Effect of LLM exposure on occupation-level job postings, job-role by time fixed effects

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	-0.049*** (0.018)	-556.171*** (141.990)	-0.011*** (0.003)	-105.062 (110.474)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Jobrole-by-period fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.963	0.951	0.955	0.736

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, using job-category (1 out of 7) by year-by-month fixed effects instead of just year-by-month fixed effects. All other definitions are as in Table 4.

A3.2 Within-Firm Analysis of Seniority

Table A10: Effect of LLM exposure on firm-level job outcomes, within firm effects

LLM exposure x Post x High seniority	−0.002 (0.002)
LLM exposure x Post x Low seniority	−0.007*** (0.001)
Firm-by-period FE	Yes
Firm-seniority FE	Yes
Seniority-by-period FE	Yes
Observations	3,792,846
Adjusted R ²	0.992

Notes: This table reports triple-difference estimates of the effect of LLM exposure on employment by seniority level within firms. The dependent variable is log employment at the firm-seniority-month level. We interact the post-ChatGPT indicator and firm-level LLM exposure with seniority indicators, using mid-seniority positions as the reference category. Low seniority corresponds to levels 1–2, mid seniority to levels 3–4, and high seniority to levels 5–7. All regressions include firm-by-period fixed effects, firm-by-seniority fixed effects, and seniority-by-period fixed effects. All other definitions are as in Table 3.

A3.3 Firm Selection

Table A11: Effect of LLM exposure on firm-level job outcomes, min. 10 employees

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
LLM exposure x Post	0.0004 (0.001)	−0.002*** (0.001)	0.002*** (0.001)	−0.001*** (0.0002)	160.109*** (52.656)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	11,476,930	11,307,751	10,759,217	11,307,751	11,476,930
Adj. R-squared	0.973	0.967	0.968	0.941	0.944

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes. We use all firms with at least 10 listed employees in October 2022, instead of 100 employees as in our main analysis. All other definitions are as in Table 3.

Table A12: Effect of LLM exposure on firm-level job outcomes, min 50 employees

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
LLM exposure x Post	−0.003** (0.001)	−0.004*** (0.001)	−0.002 (0.002)	−0.0004** (0.0002)	136.552** (53.014)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	2,541,572	2,541,147	2,530,632	2,541,147	2,541,572
Adj. R-squared	0.98	0.978	0.977	0.969	0.966

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes. We use all firms with at least 50 listed employees in October 2022, instead of 100 employees as in our main analysis. All other definitions are as in Table 3.

Table A13: Effect of LLM exposure on firm-level job listings, different minimum firm size thresholds

	min. 10 employees	min. 50 employees
LLM exposure x Post	−0.004*** (0.0003)	−0.008*** (0.001)
Firm fixed-effects	Yes	Yes
Industry-by-period fixed effects	Yes	Yes
Observations	11,476,930	2,541,572
Adjusted R ²	0.487	0.504

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity. We use all firms with at least 10 (column 1) or 50 (column 2) listed employees in October 2022, instead of 100 employees as in our main analysis. All other definitions are as in Table A1.

A3.4 Alternative Task-Importance Weights

Table A14: Effect of LLM exposure on firm-level job outcomes, onet weights

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	−0.002 (0.002)	−0.004* (0.002)	−0.001 (0.002)	−0.001*** (0.0003)	98.318* (56.416)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.983	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, using ONET importance weights for tasks instead of the weights used by Eloundou et al. (2024). All other definitions are as in Table 3.

Table A15: Effect of LLM exposure on firm-level job listings, onet weights

LLM exposure x Post	−0.010*** (0.002)
Firm fixed-effects	Yes
Industry-by-period fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.522

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job postings, using ONET importance weights for tasks instead of the weights used by Eloundou et al. (2024). All other definitions are as in Table A1.

Table A16: Effect of LLM exposure on occupation-level job outcomes, onet weights

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	−0.040** (0.018)	−352.827*** (128.078)	−0.007*** (0.002)	−31.785 (102.706)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.962	0.948	0.952	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job postings, using ONET importance weights for tasks instead of the weights used by Eloundou et al. (2024). All other definitions are as in Table 4.

A3.5 Weighted Regressions

Table A17: Effect of LLM exposure on firm-level job outcomes, weighted regressions

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	−0.003 (0.002)	−0.006*** (0.002)	−0.0001 (0.003)	−0.002*** (0.0004)	220.367*** (54.378)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.998	0.998	0.998	0.991	0.991

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, weighting observations by pre-treatment employment (October 2022). All other definitions are as in Table 3.